

ReDataX

M4: Direct Value Regression Model

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1 Model role and testable hypothesis

Models M0–M3 solve a binary-classification problem: they estimate the probability that aggressor-aligned markout will be positive, but they do not distinguish events by the size of the movement. That is not quite enough for a decision. A high probability of a weak move may be worth less than a lower probability of a large move. M4 therefore makes its prediction in basis points, opening the door to a more economically informed decision.

Hypothesis 1.1. An estimate that accounts for both the occurrence and magnitude of positive markout makes it possible to identify a subset of notional for which protected value exceeds the cost of action in subsequent periods.

The hypothesis is tested for M4 together with the new P2 policy.

2 Why move to a direct value estimate?

A protective action, such as hedging, has a cost. The platform’s benefit, meanwhile, depends on markout magnitude: the larger the adverse price move, the more value a timely protective position can preserve. Regressing positive markout makes it possible to rank decision times by expected benefit and allocate a limited intervention budget more efficiently.

3 Data and target variable

The data source is Binance Spot aggregate trades (`aggTrades`) for BTCUSDT, ETHUSDT, and ETHBTC. Trades are placed on a one-second UTC grid. For every second, the pipeline constructs signed flow, quote volume, and volume-weighted average price $P_s(t)$. The orientation of the current second is $d_A(t) = \text{sign } \phi_A(t)$.

Definition 3.1. The *aggressor-aligned markout* at horizon H is

$$m_{A,H}(t) = 10^4 d_A(t) \frac{P_A(t+H) - P_A(t)}{P_A(t)}.$$

Definition 3.2. The M4 label is the positive part of aggressor-aligned markout:

$$S_{A,H}(t) = \max\{m_{A,H}(t), 0\} \quad (\text{in basis points}).$$

During training, the target is capped at $C = 100$ bps and log-transformed:

$$y^{\text{M4}}(t) = \log(1 + \min\{S_{A,H}(t), C\}).$$

The forecast horizon is $H = 600$ seconds and the decision stride is 10 seconds. An observation is admissible when at least 600 seconds have elapsed since the start of the day; the target market is active at time t and $d_A(t) \neq 0$; current and future VWAP values are finite and positive; notional is positive; and $t + H$ remains within the same day. The future price is used to construct the target and never enters the feature vector.

4 Feature space

M4 and M5 use the same 208-column feature matrix built from three markets on a common one-second grid. It contains current log-volume for each market; target-oriented flow, absolute value, square, and fourth power of the smoothed field at 1, 2, 4, 8, 16, 32, and 64 seconds (84 features); oriented and absolute returns over 5, 10, 30, 60, 120, 300, and 600 seconds (42 features); realized one-second volatility over the same windows (18 features); oriented imbalance and log cumulative volume over windows from 10 to 600 seconds (36 features); cross-scale flow acceleration (9 features); residual change (14 features); and sine and cosine of time of day (2 features).

5 Statistical model and training

The current M4 implementation uses gradient boosting with squared-error loss.

The estimator is `HistGradientBoostingRegressor`. The selected final model uses the `medium` preset:

learning rate = 0.05, number of iterations = 160, max leaf nodes = 31,
min samples per leaf = 60, L_2 regularization = 1, maximum number of bins = 63.

Observations are weighted in proportion to the fourth root of target-market notional:

$$w_i = \text{clip} \left[\left(\frac{N_i}{\text{median}(N_{\text{train}})} \right)^{0.25}, 0.25, 4 \right].$$

This gives larger trades more influence without allowing a handful of outliers, or fluctuations, to dominate the loss. Predictions on the log scale are transformed back into basis points as

$$\hat{S}_i^{\text{M4}} = \text{clip} \{ \exp(\hat{y}_i) - 1, 0, C \}.$$

Remark 5.1. Because the transformation is nonlinear, $\exp(\mathbb{E}[\log(1 + S) \mid X]) - 1$ is generally not equal to $\mathbb{E}[S \mid X]$. No retransformation-bias correction is currently applied.

The development search covers horizons of 120, 300, and 600 seconds; `compact` and `medium` presets; budgets of 1%, 2%, 5%, and 10%; safety margins of 0, 0.05, and 0.10 bps; probability thresholds of 0, 40%, 50%, and 60%; and multipliers $\lambda \in \{1, 1.25, 1.5\}$. A candidate is admissible if mean daily value is positive, $R = \bar{V}_{\text{day}} - 0.5 \text{ sd}(V_{\text{day}}) > 0$, and at least five of seven days have positive value. Once the horizon is fixed on the development sample, the model is refitted on the first 35 days and the final seven days are opened only once.

6 Results

The final specification uses a 600-second horizon and the `medium` preset. It is evaluated under policy P2.

During the preceding development week, M4 achieved mean daily values of 11.47 for BTCUSDT and 25.23 for ETHUSDT, with positive-day shares of 85.7% and 100%, respectively.

Table 1: Final period, 10–16 February 2025 (P2 policy based on M4).

Metric	BTCUSDT	ETHUSDT
Pooled net value, USDT per 1m USDT	7.20	23.06
Mean daily value, USDT per 1m USDT	4.71	16.58
Daily standard deviation	8.66	18.39
Share of positive days	71.4%	100.0%
Notional touched	2.98%	6.97%
Exposure captured	8.39%	14.45%
Risk concentration	2.82	2.07
Benefit-cost ratio	5.84	7.62

Result 6.1. M4 confirms that a magnitude-aware markout forecast can produce positive value in both final-test markets. The model is particularly efficient per unit of capital touched, although its absolute value is lower than that of M5 for both markets.

7 Limitations and reproducibility

M4 inherits the limitations of the preceding models and introduces several new ones. Public Binance flow does not reproduce the client flow or execution process of a particular institution. The parameters ι , η , and c come from the literature rather than being estimated from bank data; changing any one of them shifts the break-even threshold. The target is capped at 100 bps, and the inverse log transformation is not corrected for bias. The final period contains only seven calendar days, limiting what can be concluded about tail risk and robustness to changes in market regime.

References

- [1] L. R. Glosten, P. R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100, 1985.
- [2] D. Easley, M. M. López de Prado, M. O’Hara. Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5), 1457–1493, 2012.
- [3] R. Cont, A. Kukanov, S. Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1), 47–88, 2014.
- [4] J. H. Friedman. Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232, 2001.
- [5] N. Duan. Smearing estimate: a nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383), 605–610, 1983.
- [6] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830, 2011.
- [7] Binance. Spot REST API: Compressed/Aggregate Trades List. <https://developers.binance.com/en/docs/catalog/core-trading-spot-trading/api/rest-api/market>.